Sahaj Bhadja

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Summary

Software Engineer with 2+ years in **C++** and Python on **Linux**, building concurrent systems, time-series data pipelines, and motion-planning software validated in simulation, with Git, **CI/CD**, and automated testing.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built a fault-tolerant **concurrent** processing system with explicit state transitions and failure isolation, handling 1K+ jobs at 3x throughput without cascading failures under heavy concurrent production load in a live deployment.
- Built data pipelines with validation and reconciliation checks, shipped through **CI/CD-backed releases**, sustaining 99%+ reliability over 5K+ monthly jobs while troubleshooting failures from logs, test results, and data.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Designed a **multithreaded** batch-processing engine with a thread pool and fault-tolerant execution, processing 1K+ files per run while keeping backend services stable for 150+ users at peak load.
- **Debugged** timeouts and failures across external model providers by adding fallback routing and timeout handling, keeping 10K+ daily requests stable and cutting instructor feedback turnaround 60% across 10K+ submissions.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built Python pipelines that process 100K+ vehicle trajectory logs sampled at 10Hz, aligning ego state and nearby agents into a **global coordinate frame** for training and evaluating a motion planner.
- Validated the planner and a map-aware encoder against three baselines in **closed-loop simulation**, analyzing failures such as off-road trajectories and improving long-horizon planning 18% and closed-loop stability 15% overall.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Maintained **automated build and test** pipelines that compiled code nightly, installed executables on Linux, Windows, and macOS machines, and ran 20+ unit and integration suites with coverage reports.
- Used **Git** feature, staging, and production branches with Jenkins, and built object-oriented Factory and Singleton modules, improving response times 30% and cutting technical debt 20% in Agile teams.

Pandit Deendayal Energy University **January 2023 – May 2024**
Research Fellowship *India*

- Designed a quantum-resistant blockchain protocol integrating 5+ lattice-based **cryptographic** primitives, analyzing consensus mechanisms to develop a novel algorithm resilient to centralized control and quantum attacks on signatures.
- **Benchmarked** the design across 100 to 1,000 transaction workloads, measuring 957.8 TPS at 224ms latency against a baseline, and presented the work at IISc Bengaluru in an IEEE publication.

Skills

Languages: C++, Python, Java, SQL, Bash, Shell, JavaScript
Systems: Linux, Operating Systems, Concurrency, Multithreading, CPU Scheduling, Data Structures, Algorithms, OOP
Algorithms & Data: Simulation, Coordinate Transforms, Time-Series Processing, PyTorch, Reinforcement Learning, Cryptography
Quality: Unit Testing, Integration Testing, Code Coverage, PyTest, Mockito, Debugging, Performance Optimization
Tools: Git, GitHub Actions, Jenkins, CI/CD, Docker, TCP/IP, Jira, Agile, Windows, macOS, PostgreSQL, REST APIs

Projects

GNSS-INS Fusion Navigator | *C++17, Eigen, CMake, GoogleTest, Linux*

- Designed a **C++17** error-state Kalman filter that fuses IMU and GNSS measurements, keeps navigating on inertial data through simulated GPS outages, and is verified by **unit tests** on recorded trajectories.

Navigation Sensor Interface Layer | *C++17, Serial/UART, Ethernet, GitLab CI, Python*

- Designed a C++ driver layer that parses IMU and GNSS receiver messages over serial and Ethernet into timestamped measurements, with **GitLab CI** builds, static analysis, and Python **replay tests**.

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2, OS Scheduling*

- Implemented FCFS, SJF, SRT, Round Robin, priority, and multilevel queue **scheduling** in C++ with Gantt visualization, testing 1K+ process combinations and cutting average turnaround time 25% across test scenarios.